

EDUCATION

University of Kentucky
Bachelor of Science in Aerospace Engineering
GPA 3.5/4.0 Graduation date: May 8, 2026

TECHNICAL / NON-TECHNICAL SKILLS

- **CAD Software:** Inventor, OnShape, Fusion 360
 - **Programming/Simulation:** Python, MATLAB, FEA, CFD, Flight Dynamics (AVL – aircraft stability and control)
 - **Manufacturing/Project Tools:** CNC, perpendicular and parallel machining, Mission planners, G-code, Microsoft Office applications
 - **Professional Skills:** Technical writing, presentations, project management
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EXPERIENCE

Detailer/Driver, GoRentals, Lexington KY

July 2025 – September 2025

- Detailed and delivered rental vehicles for high priority people at Bluegrass airport. Worked with the likes of NASA, political figures, and celebrities.
- Gained experience in airport operations, attention to detail, and customer service.

Research Assistant, Materials Lab, University of Kentucky

June 2024 - July 2024

- Polished machining tools and took data on stress due to perpendicular and parallel machining.
- Gained experience in CNC machining and how it creates stress as well as polishing tools.

Assembler/CNC & Laser operator, Emblem Pros, Lexington KY

October 2021 – December 2023

- Vehicle emblem assembly via Ski co. LLC, authorized by General Motors.
 - Laser cut metal, plastics, filed and assembled custom GM emblems per customer request.
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PROJECTS

Hybrid Turbine Airfoil, Aerospace Engineering Capstone Team Project

August 2025 - May 2026

- Collaborated on a 6-person team to design internal structures for a low-pressure turbine blade in a jet engine
- Developed and evaluated 6 unique internal structural concepts under thermal, structural, and modal loading conditions
- Designed and built a thermal test setup to replicate low-pressure turbine operating environments
- Conducted 60 experimental tests to assess thermal performance and identify the most robust design
- Analyzed test data to validate design effectiveness and support final design selection

Supersonic Nozzle Validation: ANSYS Fluent Analysis, CFD Project

March 2026 - May 2026

- Validated a 2D converging–diverging nozzle CFD model in ANSYS Fluent against 1D isentropic theory, achieving 0.2% error in exit Mach number and <0.01% mass flow deviation
- Implemented a density-based implicit solver with Roe flux-difference splitting and second-order discretization to accurately capture Prandtl–Meyer expansion waves and overexpanded flow behavior
- Performed a three-level grid independence study (1.2k to 20k cells), reducing discretization error to 0.06% and confirming solution convergence
- Computed net thrust by analyzing momentum flux and pressure differentials, matching theoretical predictions within 1.1% (5,297 N)

Lunar Nuclear Power Plant, Systems Engineering Team Project

August 2025 - December 2025

- Partnered with a five-member team to design a fission-based lunar surface power system with an operational life exceeding ten years
- Defined functional and performance requirements, ConOps diagrams, and event timelines while integrating mission, environmental, operational, and regulatory design factors
- Developed an initial segmented architecture mapping top-level functions to power conversion, heat rejection, and distribution subsystems with early conceptual and physical configurations
- Performed trade studies and risk assessments by evaluating Technology Readiness Levels, performance risks, and mitigation strategies in support of NASA's surface power initiatives
- Delivered a final design package with architecture layers and N2 interface diagrams to demonstrate system-level engineering processes for mission-critical space power systems

Aircraft Modeling & Simulation, Flight Dynamics Team Project

February 2025 - May 2025

- Created a personal transport aircraft concept in Inventor that met lateral and longitudinal stability requirements
- Modeled aerodynamic forces and moments in AVL to generate data for system identification and dynamic stability analysis
- Developed and linearized a state-space model from AVL outputs to analyze more than five flight modes under varying conditions
- Simulated initial condition and forced responses in MATLAB while evaluating two flight controllers to enhance stability and handling qualities

Numerical Methods for Airfoil Analysis, Engineering Analysis Project

October 2024 - December 2024

- Developed a MATLAB integration script to compute lift and centroid location from discrete airfoil pressure data
- Analyzed over one thousand pressure measurements to generate accurate aerodynamic force predictions and performance plots
- Verified results against theoretical models, maintaining error margins under three percent across test cases
- Presented a detailed code review to peers and faculty, highlighting both technical accuracy and clear communication

Spar Design, Aerospace Structures Team Project

August 2024 - December 2024

- Worked with a five-member team to design a cost-efficient wing structure using innovative materials resilient to static loads, engine-out cases, and varied takeoff conditions
- Modeled over thirty spar and engine placement configurations in MATLAB and Excel to evaluate stress distributions
- Refined the design to reduce weight by fifteen percent while maintaining compliance with FAA regulations
- Produced a final technical report highlighting trade-offs among weight, structural strength, and cost in aerospace applications

Autonomous UAV (Unmanned Aerial Vehicle), Aerospace Project

August 2023 - December 2023

- Designed, assembled, and programmed a UAV capable of autonomously completing a timed obstacle course
- Implemented ArduPilot firmware and configured flight paths in Mission Planner to refine control response
- Achieved full mission success on the first flight, surpassing peer teams and earning top performance evaluations
- Gained practical experience in UAV integration, embedded programming, and real-time flight testing

Personal Projects

Back Deck Demolition & Remodel,

May 2025 - July 2025

- Demolished and rebuilt failing back deck
- Applied principles of structural statics and load distribution to redesign the deck framework, ensuring stability and resilience under dynamic and long-term loading conditions
- Integrated hand-tool precision and mechanical assembly techniques, translating theoretical understanding of fastener mechanics into accurate, reliable construction methods
- Strengthened expertise in practical application of theoretical mechanics, bridging the gap between classroom concepts and real-world engineering challenges

Vehicle Modification

December 2020 – Present

- Successfully modified many components of my 2013 Scion FR-S vehicle
- Replaced factory air intake with a high performing cold air intake, replaced the exhaust from the cat-back, replaced original suspension parts with coil overs, replaced rotors and brake pads, flashed the ECU with an Open flash tablet (OFT tuning) to run on E85 gas, and more.
- Continuously improving critical thinking and hands on work outside of engineering calculations.

ACTIVITIES/AWARDS

- Offered a position in UKY's competitive dual MBA/Engineering Masters Program
- UKY American Institute of Aeronautics and Astronautics (AIAA) member
- UKY Provost Scholarship recipient
- UKY Dean's List recipient (Fall 2022, Spring 2023, Fall 2024)
- 3.4+ GPA every semester
- Consistently workout, focused on weightlifting